MGM'S College of Engineering, Nanded
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

"NEWSPULSE: AI-POWERED FAKE NEWS DETECTION"

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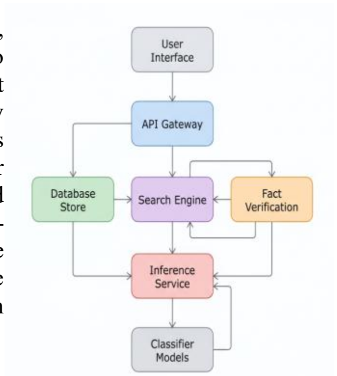
Introduction: The exponential growth of digital media has accelerated the spread of misinformation, highlighting the need for real-time verification platforms. NewsPulse addresses this challenge by combining robust deep-learning text classification with dynamic, real-world factual validation. By merging stylistic language processing with external web searching and large language model validation, the platform provides highly accurate, explainable fake news detection.

System Architecture:

NewsPulse leverages a decoupled, multi-tier microservices architecture to process data efficiently without introducing network latency bottlenecks. An asynchronous Node.js and Express gateway intercepts user payloads, handles secure sessions, and passes verified strings to a high-performance FastAPI tier. The computational core handles intensive model predictions in absolute isolation from frontend operations.



Methodology:

The methodology begins with cleaning and normalizing input text, followed by TF-IDF and dense token feature extraction. Clean inputs are processed in parallel through a dual classification framework utilizing Logistic Regression and deep RoBERTa models. Concurrently, a retrieval-augmented validation protocol queries web search results, passing this live context to Google Gemini to decide whether to execute a dynamic state override on the classification.

Conclusion: In conclusion, NewsPulse demonstrates the immense value of combining offline stylistic analysis with live, retrieval-augmented fact verification. The microservices-based design successfully isolates intensive tensor operations from client actions, ensuring high scalability and throughput. Moving forward, the framework can be extended to support multimodal content, multilingual analysis, and automated URL credibility scoring

Maps to:
PO1 to PO5, PO9 , PO10, PO12
PSO1 to PSO3

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